

Phase 1 Comprehensive Sustainability Methodology and District Test Packet

This document is designed as a parser-aware stress test for the Istanbul sustainability reporting pipeline. It mixes methodology text, factor metadata, narrative facts, and structured district tables in one PDF so the uploaded file can exercise multiple system behaviours in a single run.

Expected behaviour: the system should extract a custom formula, override electricity and natural gas factors, recognise water as a sustainability KPI, and generate a report with district-level water analysis.

1. Carbon Methodology

Total CO2 = electricity * electricity_factor + gas * gas_factor - tree_count * tree_credit + direct_emissions
tree_credit = 0.08

Water is tracked as a resource KPI. In the default methodology below, water is reported for efficiency and trend analysis, but it is not converted into carbon unless a future methodology explicitly defines a separate water factor and formula.

2. Factor Metadata Table

Parameter	Value
Electricity emission factor	Approved factor for 2024: 0.45 kgCO2e/kWh
Natural gas emission factor	Verified methodological value: 2.04 kgCO2e/m3
Tree credit	Tree credit constant for district green assets: 0.08
Water note	Water values are reported in m3 and remain a resource KPI in the default model.

3. Narrative District Facts

In 2024, Kadikoy recorded 3,783,227 kWh electricity consumption, 311,829 m3 natural gas consumption, 898,803 m3 water use, 11,038 trees in the monitored sink inventory, and 573 tCO2e direct emissions.

In 2024, Tuzla recorded 2,730,625 kWh electricity consumption, 232,260 m3 natural gas consumption, 649,700 m3 water use, 7,712 trees in the monitored sink inventory, and 439 tCO2e direct emissions.

In 2024, Sile recorded 458,323 kWh electricity consumption, 40,072 m3 natural gas consumption, 92,666 m3 water use, 1,552 trees in the monitored sink inventory, and 55 tCO2e direct emissions.

In 2024, Esenyurt recorded 7,830,459 kWh electricity consumption, 863,358 m3 natural gas consumption, 1,967,653 m3 water use, 24,382 trees in the monitored sink inventory, and 1,254 tCO2e direct emissions.

In 2024, Besiktas recorded 1,368,633 kWh electricity consumption, 102,854 m3 natural gas consumption, 343,402 m3 water use, 4,549 trees in the monitored sink inventory, and 232 tCO2e direct emissions.

4. Core District Energy and Water Dataset (2023-2024)

This table is the primary structured district dataset. It provides energy and water values for all 39 districts.

District	Year	Electricity kWh	Natural Gas m3	Water m3
Esenyurt	2023	7,750,146	637,479	1,817,068
Esenyurt	2024	8,253,905	685,290	1,962,433
Kucukcekmece	2023	6,392,097	541,836	1,492,013
Kucukcekmece	2024	6,903,465	590,601	1,633,754
Pendik	2023	6,437,402	560,264	1,496,545
Pendik	2024	6,759,272	593,880	1,593,820
Umraniye	2023	6,545,638	583,130	1,253,730
Umraniye	2024	6,971,104	626,865	1,354,028
Bagcilar	2023	5,165,735	605,027	1,280,819
Bagcilar	2024	5,578,994	659,479	1,402,497
Bahcelievler	2023	4,161,350	312,640	1,024,166
Bahcelievler	2024	4,369,418	331,398	1,090,737
Basaksehir	2023	4,369,527	340,866	1,068,226
Basaksehir	2024	4,238,441	334,049	1,052,203
Sultangazi	2023	4,530,859	365,221	910,406
Sultangazi	2024	4,893,328	398,091	996,895
Maltepe	2023	4,714,895	391,157	950,329
Maltepe	2024	4,950,640	414,626	1,012,100
Uskudar	2023	3,754,346	411,435	977,158
Uskudar	2024	3,998,378	442,293	1,055,331
Sancaktepe	2023	3,917,900	433,912	1,009,925
Sancaktepe	2024	4,231,332	472,964	1,105,868
Gaziosmanpasa	2023	3,894,135	277,469	822,839
Gaziosmanpasa	2024	4,088,842	294,117	876,324
Kartal	2023	4,071,392	302,025	860,890
Kartal	2024	4,336,032	324,677	929,761
Kadikoy	2023	4,117,985	316,415	871,288
Kadikoy	2024	3,994,445	310,087	858,219
Kagithane	2023	3,258,866	332,582	888,375
Kagithane	2024	3,421,809	352,537	946,119
Avcilar	2023	3,401,918	352,530	757,940
Avcilar	2024	3,623,043	378,970	818,575
Eyupsultan	2023	3,461,258	363,559	769,640
Eyupsultan	2024	3,738,159	396,279	842,756
Beylikduzu	2023	3,620,777	245,333	803,677
Beylikduzu	2024	3,801,816	260,053	855,916
Esenler	2023	3,770,504	266,622	835,557
Esenler	2024	4,015,587	286,619	902,402

District	Year	Electricity kWh	Natural Gas m3	Water m3
Atasehir	2023	3,008,512	284,366	708,855
Atasehir	2024	3,249,193	309,959	776,196
Sultanbeyli	2023	2,925,169	282,286	685,823
Sultanbeyli	2024	2,837,414	276,640	675,536
Arnavutkoy	2023	2,917,937	286,775	681,091
Arnavutkoy	2024	3,107,603	308,283	735,578
Fatih	2023	3,011,288	300,777	700,054
Fatih	2024	3,252,191	327,847	766,559
Sariyer	2023	3,097,049	200,032	593,198
Sariyer	2024	3,251,901	212,034	631,756
Cekmekoy	2023	2,306,500	200,633	571,885
Cekmekoy	2024	2,456,422	215,680	617,636
Tuzla	2023	2,423,037	216,566	596,343
Tuzla	2024	2,616,880	236,057	652,996
Buyukcekmece	2023	2,305,565	211,013	563,645
Buyukcekmece	2024	2,420,843	223,674	600,282
Zeytinburnu	2023	2,358,031	220,376	473,810
Zeytinburnu	2024	2,287,290	215,968	466,703
Bayrampasa	2023	2,451,342	233,396	494,090
Bayrampasa	2024	2,647,449	254,402	541,029
Sisli	2023	1,912,300	151,936	497,722
Sisli	2024	2,007,915	161,052	530,074
Gungoren	2023	1,939,588	159,538	499,971
Gungoren	2024	2,065,661	171,503	539,969
Beykoz	2023	2,009,220	170,314	424,552
Beykoz	2024	2,169,958	185,642	464,884
Silivri	2023	2,054,648	178,821	434,452
Silivri	2024	2,157,380	189,550	462,691
Bakirkoy	2023	1,959,471	174,563	414,587
Bakirkoy	2024	2,086,837	187,655	447,754
Beyoglu	2023	1,576,734	184,672	429,822
Beyoglu	2024	1,529,432	180,979	423,375
Besiktas	2023	1,280,709	96,219	285,339
Besiktas	2024	1,344,744	101,992	303,886
Catalca	2023	660,504	51,525	146,868
Catalca	2024	703,437	55,389	158,617
Sile	2023	428,770	34,562	95,171
Sile	2024	463,072	37,673	104,212
Adalar	2023	157,051	13,029	34,803
Adalar	2024	164,904	13,811	37,065

5. Sink Inventory and Direct Emissions Dataset (2023-2024)

This companion table provides the tree_count sink input and district-level direct emissions used by the custom formula.

District	Year	Tree Count	Direct tCO2e
Esenyurt	2023	22,404	963
Esenyurt	2024	22,754	1,030
Kucukcekmece	2023	17,640	840
Kucukcekmece	2024	17,990	911
Pendik	2023	16,996	887
Pendik	2024	17,346	936
Umraniye	2023	16,578	619
Umraniye	2024	16,928	662
Bagcilar	2023	16,200	679
Bagcilar	2024	16,550	737
Bahcelievler	2023	12,548	576
Bahcelievler	2024	12,898	608
Basaksehir	2023	12,593	633
Basaksehir	2024	12,943	617
Sultangazi	2023	12,522	449
Sultangazi	2024	12,872	487
Maltepe	2023	12,522	504
Maltepe	2024	12,872	532
Uskudar	2023	12,378	550
Uskudar	2024	12,728	588
Sancaktepe	2023	12,322	598
Sancaktepe	2024	12,672	649
Gaziosmanpasa	2023	11,771	406
Gaziosmanpasa	2024	12,121	428
Kartal	2023	11,804	456
Kartal	2024	12,154	488
Kadikoy	2023	11,520	490
Kadikoy	2024	11,870	478
Kagithane	2023	11,345	526
Kagithane	2024	11,695	555
Avcilar	2023	11,312	374
Avcilar	2024	11,662	400
Eyupsultan	2023	11,064	408
Eyupsultan	2024	11,414	443
Beylikduzu	2023	11,109	452
Beylikduzu	2024	11,459	477
Esenler	2023	11,135	495

District	Year	Tree Count	Direct tCO2e
Esenler	2024	11,485	530
Atasehir	2023	11,058	350
Atasehir	2024	11,408	380
Sultanbeyli	2023	10,415	363
Sultanbeyli	2024	10,765	354
Arnavutkoy	2023	10,055	383
Arnavutkoy	2024	10,405	410
Fatih	2023	10,002	415
Fatih	2024	10,352	450
Sariyer	2023	9,944	293
Sariyer	2024	10,294	309
Cekmekoy	2023	9,396	303
Cekmekoy	2024	9,746	324
Tuzla	2023	9,444	335
Tuzla	2024	9,794	363
Buyukcekmece	2023	8,859	334
Buyukcekmece	2024	9,209	352
Zeytinburnu	2023	8,781	234
Zeytinburnu	2024	9,131	228
Bayrampasa	2023	8,821	262
Bayrampasa	2024	9,171	284
Sisli	2023	8,671	280
Sisli	2024	9,021	295
Gungoren	2023	8,528	296
Gungoren	2024	8,878	317
Beykoz	2023	8,525	209
Beykoz	2024	8,875	227
Silivri	2023	8,468	230
Silivri	2024	8,818	243
Bakirkoy	2023	8,078	233
Bakirkoy	2024	8,428	249
Beyoglu	2023	8,124	254
Beyoglu	2024	8,474	248
Besiktas	2023	7,106	141
Besiktas	2024	7,456	149
Catalca	2023	5,318	77
Catalca	2024	5,668	82
Sile	2023	4,723	53
Sile	2024	5,073	58
Adalar	2023	4,093	20
Adalar	2024	4,443	21

6. Supplementary Resource Metrics (Optional Second Ring)

This appendix introduces optional resource metrics such as wastewater, solid waste, diesel, gasoline, and general fuel rows.

District	Year	Metric	Unit	Value
Kadikoy	2024	Wastewater Treated	m3	1,180,000
Kadikoy	2024	Solid Waste	tonnes	182,500
Tuzla	2024	Diesel Consumption	L	88,400
Pendik	2024	Fuel Consumption	L	125,900
Besiktas	2024	Gasoline Consumption	L	42,300
Sile	2024	Wastewater Flow	m3	96,800
Esenyurt	2024	Municipal Waste	tonnes	264,000
Uskudar	2024	Wastewater Treated	m3	920,000
Bakirkoy	2024	Solid Waste	tonnes	98,600
Sariyer	2024	Diesel Consumption	L	53,200

7. Validation Checklist

- The PDF should extract a custom carbon formula with tree_count and direct_emissions bound from uploaded data.
- The electricity factor should resolve to 0.45 and the natural gas factor should resolve to 2.04 instead of the defaults.
- The report should include district-level water values and a Water Overview section.
- The document should remain valid even though water is not part of the default carbon formula.
- Optional resource metrics should be available for exploratory parser checks.